

Module 37 - Array Methods (splice, pop, push, shift, unshift)

1. What are Array Methods?

Array methods are built-in JavaScript functions used to perform different operations on arrays, such as adding, removing, searching, modifying, sorting, and processing elements.

2. pop() Method

The pop() method is used to remove the last element from an array.

Example Code:

```
let arr = [100, 200, 300, 400];  
// Removes the last element from the array and returns it.  
let poppedVal = arr.pop();  
console.log(poppedVal, arr);
```

3. push() Method

The push() method is used to add one or more elements to the end of an array.

Example Code:

```
// Adds one or more elements to the end of the array.  
arr.push(500, 550, 600);  
console.log(arr);
```

4. shift() Method

The shift() method is used to remove the first element from an array.

Example Code:

```
// Removes the first element from the array and returns it.  
let shiftedVal = arr.shift();  
console.log(shiftedVal, arr);
```

5. unshift() Method

The unshift() method is used to add one or more elements to the beginning of an array.

Example Code:

```
// Adds one or more elements to the beginning of the array.  
arr.unshift(50, 75);  
console.log(arr);
```

6. splice() Method

The splice() method is used to add, remove, or replace elements at a specific index in an array.

Example Code:

```
// Syntax: array.splice(startIndex, deleteCount, item1, item2, ...)

// Delete using splice
// Removes 2 elements starting from index 2 and returns them.
let removedItems = arr.splice(2, 2);
console.log(removedItems, arr);

// Insert using splice
// Inserts 250 and 350 at index 2 without removing any elements.
arr.splice(2, 0, 250, 350);
console.log(arr);

// Replace using splice
// Replaces 2 elements starting from index 4 with 800 and 900.
arr.splice(4, 2, 800, 900);
console.log(arr);
```